Heat and Temperature Investigation

Aim: To observe the relationship between heat and temperature.

Equipment

- Two 250cm^3 beakers
- Thermometer
- Test tube with ice (sealed with a rubber bung)
- Test tube with wax (sealed with a rubber bung)
- 10ml syringe
- Hot water
- Test tube rack

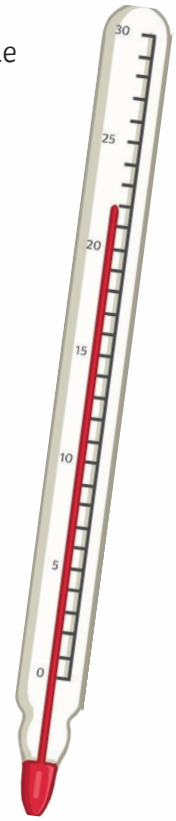
Safety Information

Glassware - risk of cuts if broken.
Apply pressure to wounds and inform teacher immediately.

Hot water – risk of scalds or burns
Rinse with cold running water for 10 minutes.

Method

1. Fill one of the beakers approx $\frac{3}{4}$ full with hot water.
2. Place the thermometer into the hot water and record the initial temperature. Leave the thermometer in the beaker for the duration of the investigation.
3. Draw 5ml air into the syringe and place into the beaker of hot water.
4. Observe any changes to the water temperature or syringe. Record these into your results table.
5. Remove the syringe and place the test tube containing ice into the beaker.
6. Observe any changes to the water temperature or ice. Record these into your results table.
7. Remove the test tube and place the test tube containing wax into the beaker.
8. Observe any changes to the water temperature or wax. Record these into your results table.
9. Remove the test tube and pour 100ml of cold water into the beaker.
10. Observe any changes to the water temperature. Record these into your results table.



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Remember:

Heat is the amount of thermal energy stored, and is measured in joules (J).

Temperature is a measure of how hot or cold something is, and is measured in degrees Celsius (˚C).

Results

	Start temperature, ˚C	End temperature, ˚C	Observation notes
Syringe of air into hot water			
Tube of ice in hot water			
Tube of wax in hot water			
Cold water poured into hot water			

Questions

1. Using your observations, describe what happens to a solid object (ice / wax) when it is heated?
2. Using your observations, describe what happens to a liquid object (inside thermometer) when it is heated?
3. Using your observations, describe what happens to a gas object (inside syringe) when it is heated?

Extra Challenge

4. The temperature of a lake is 15˚C. The temperature of a mug of cocoa is 85˚C.
   a. Which has the highest temperature?
   b. Which has the largest thermal energy store?

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Answers

Results

	Start temperature, ˚C	End temperature, ˚C	Observation notes
Syringe of air into hot water		Decreasing	Gas expands (syringe moves up), bubbles
Tube of ice in hot water		Decreasing	Ice melts (turns liquid)
Tube of wax in hot water		Decreasing	Wax melts (turns liquid)
Cold water poured into hot water		Decreasing	

Questions

1. Using your observations, describe what happens to a solid object (ice / wax) when it is heated?
   Melts

2. Using your observations, describe what happens to a liquid object (inside thermometer) when it is heated?
   Expands

3. Using your observations, describe what happens to a gas object (inside syringe) when it is heated?
   Expands

Extra Challenge

4. The temperature of a lake is 15˚C. The temperature of a mug of cocoa is 85˚C.
   a. Which has the highest temperature?
      Cocoa

   b. Which has the largest thermal energy store?
      Lake